Problem Set 7 – Multiple Regression Part IV

Problem 1 (20 points)

Please access the *Real Estate Data* dataset. These data are from Kaggle, and no license agreement was listed as known. Latitude, longitude pairs within the data suggest they correspond to housing prices in New Taipei City, in the Xindian District, Taiwan. This led to certain (perhaps incorrect) assumptions being made about the units for the variables. The dataset consists of:

- Age of House (years)
- Distance to Nearest MRT Station (meters)
- Number of Convenience Stores (within a kilometer)
- Latitude
- Longitude
- House Price Per Unit Area (Taiwanese dollars)

a. (12 points) We used JMP's Fit Model to fit House Price Per Unit Area to nine polynomial models with the regressor Latitude (line, parabola, quadratic, etc.). Use "Macro to Degree" to make sure you get the full polynomial model in each case. For each model fit, we recorded the R^2, Press R^2, adjR^2 and AICc values. We have left four entries unfilled. Please fill those in below[1]. You are advised to first make sure you can recreate a given value so that you know you are running the right models.

Model Degree	R^2	PRESS R^2	AdjR^2	AICc
1	0.2985	0.2907	0.2967	3194.723
2	0.3348	0.3215	0.3316	3174.709
3	0.3662	0.3046	0.3616	3156.735
4	0.3939	0.3809	0.3879	3140.346
5	0.3939	0.3250	0.3864	3142.413
6	0.3943	**-2.7393**	0.3854	3144.158
7	0.4068	-3.5032	**0.3965**	**3137.685**
8	0.4119	-7.6135	0.4003	3136.178
9	0.4136	-6.7836	0.4005	3137.103

b. (8 points) Refer to the table in part a. In 100 words or less explain whether PRESS R^2, AdjR^2 and AICc offer consistent suggestions about when the model is overfitting.l

No, PRESS R^2, adjusted R^2, and AICc do not provide consistent indications of overfitting. PRESS R^2 reaches its highest value at degree 4 and declines substantially afterward, suggesting that higher-degree models are overfitting and have poor predictive performance. However, adjusted R^2 continues to increase through degree 9, while AICc reaches its lowest value at degree 8.

[1] R^2 is of course just 1 – (SSE/SST). Press R^2 is 1 – (PRESS/SST), so it is interpreted like an R^2 value.

Therefore, adjusted R^2 and AICc favor more complex models, whereas PRESS R^2 suggests that the fourth-degree polynomial provides the best predictive performance.

Problem 2 (10 points)

Please access the *Life Expectancy* dataset. The dataset is publicly available on the Kaggle website, and it was collected from the World Health Organization and United Nations website with the help of Deeksha Russell and Duan Wang. For this problem we are only interested in the following variables. They should all be treated as numeric and continuous for our analysis.

Response: Life expectancy (life expectancy in age)

Regressors:

1. Adult Mortality (adult mortality rates of both sexes (probability of dying between 15 and 60 years per 1000 population)
2. Infant Deaths (number of infant deaths per 1000 population)
3. Alcohol (recorded per capita -15+ - consumption, in liters of pure alcohol)
4. Percentage Expenditure (% expenditure on health as a percentage of gross domestic product per capita)
5. Hepatitis B (% HepB immunization coverage among 1-year-olds)
6. Measles (number of reported cases per 1000 population)
7. BMI (average body mass index of entire population)
8. GDP (gross domestic product per capita in USD)

a. (3 points) Use JMP to perform an all-subsets regression using the response and eight regressors shown above. Copy and paste a small part of the top of that output here so that it will be clear that you are able to get it done.

Model	Number	RSquare	RMSE	AICc	BIC	Cp	
Adult Mortality	1	0.5087	6.0493	11958.9	11975.5	859.1339	( )
BMI	1	0.2742	7.3528	12683.7	12700.3	2153.9313	( )
GDP	1	0.1917	7.7592	12883.5	12900.0	2609.0748	( )
percentage expenditure	1	0.1710	7.8580	12930.5	12947.1	2723.4833	( )
Alcohol	1	0.1242	8.0766	13032.4	13049.0	2981.6667	( )
Hepatitis B	1	0.0518	8.4040	13180.0	13196.5	3381.4976	( )
infant deaths	1	0.0337	8.4838	13215.1	13231.6	3481.3914	( )
Measles	1	0.0071	8.5998	13265.5	13282.1	3628.3251	( )
Adult Mortality,BMI	2	0.5918	5.5157	11617.0	11639.1	402.5811	( )
Adult Mortality,Alcohol	2	0.5770	5.6148	11683.1	11705.2	484.2834	( )
Adult Mortality,GDP	2	0.5722	5.6462	11703.9	11725.9	510.5207	( )
Adult Mortality,percentage expenditure	2	0.5699	5.6613	11713.8	11735.9	523.1661	( )
Adult Mortality,infant deaths	2	0.5285	5.9275	11884.4	11906.5	751.6268	( )
Adult Mortality,Hepatitis B	2	0.5252	5.9487	11897.7	11919.8	770.3270	( )
Adult Mortality,Measles	2	0.5144	6.0158	11939.3	11961.4	829.7882	( )

Model	Number	RSquare	RMSE	AICc	BIC	Cp
BMI,GDP	2	0.3710	6.8468	12419.9	12442.0	1621.5873 ( )
percentage expenditure,BMI	2	0.3619	6.8962	12446.6	12468.7	1671.7999 ( )

b. (2 points) Order all the models from smallest Cp value to largest Cp value. Copy and paste output showing the 20 models with the smallest Cp values here.

Model	Number	RSquare	RMSE	AICc	BIC	Cp
Adult Mortality, infant deaths, Alcohol, percentage expenditure, Hepatitis B, BMI, GDP	7	0.6651	5.0029	11259.6321	11309.2751	8.0326
Adult Mortality, infant deaths, Alcohol, Hepatitis B, BMI, GDP	6	0.6646	5.0051	11260.2348	11304.3706	8.6454
Adult Mortality, infant deaths, Alcohol, percentage expenditure, Hepatitis B, Measles, BMI, GDP	8	0.6652	5.0029	11260.6165	11315.7645	9.0000
Adult Mortality, infant deaths, Alcohol, Hepatitis B, Measles, BMI, GDP	7	0.6648	5.0051	11261.2479	11310.8909	9.6421
Adult Mortality, Alcohol, percentage expenditure, Hepatitis B, BMI, GDP	6	0.6629	5.0175	11269.4274	11313.5633	17.8343
Adult Mortality, Alcohol, Hepatitis B, BMI, GDP	5	0.6624	5.0200	11270.2374	11308.8639	18.6696
Adult Mortality, Alcohol, percentage expenditure, Hepatitis B, Measles, BMI, GDP	7	0.6631	5.0178	11270.6875	11320.3305	19.0734
Adult Mortality, Alcohol, Hepatitis B, Measles, BMI, GDP	6	0.6626	5.0202	11271.4441	11315.5799	19.8562
Adult Mortality, infant deaths, Alcohol, percentage expenditure, Hepatitis B, BMI	6	0.6623	5.0220	11272.7677	11316.9035	21.1844
Adult Mortality, infant deaths, Alcohol, percentage expenditure, Hepatitis B, Measles, BMI	7	0.6625	5.0220	11273.7848	11323.4278	22.1784
Adult Mortality, Alcohol, percentage expenditure, Hepatitis B, BMI	5	0.6602	5.0366	11282.5587	11321.1852	31.0763
Adult Mortality, Alcohol, percentage expenditure, Hepatitis B, Measles, BMI	6	0.6603	5.0369	11283.7913	11327.9271	32.2836
Adult Mortality, infant deaths, Alcohol, BMI, GDP	5	0.6584	5.0493	11291.9046	11330.5310	40.5419
Adult Mortality, infant deaths, Alcohol, Measles, BMI, GDP	6	0.6586	5.0495	11293.0155	11337.1513	41.6218
Adult Mortality, infant deaths, Alcohol, percentage expenditure, BMI, GDP	6	0.6586	5.0497	11293.1851	11337.3209	41.7939
Adult Mortality, infant deaths, Alcohol, percentage expenditure, Measles, BMI, GDP	7	0.6587	5.0498	11294.2847	11343.9277	42.8604

Model	Number	RSquare	RMSE	AICc	BIC	Cp
Adult Mortality, Alcohol, Measles, BMI, GDP	5	0.6547	5.0767	11311.9466	11350.5731	61.0022
Adult Mortality, infant deaths, Alcohol, percentage expenditure, BMI	5	0.6547	5.0769	11312.1261	11350.7525	61.1864
Adult Mortality, Alcohol, BMI, GDP	4	0.6542	5.0791	11312.6886	11345.8035	61.8342
Adult Mortality, infant deaths, Alcohol, percentage expenditure, Measles, BMI	6	0.6549	5.0771	11313.2717	11357.4075	62.2918

c. (5 points) Refer to your output in part b. Use the information in R^2 and Cp, along with the ever-present practical motivation to have the simplest adequate model, to suggest and defend what you'd consider the best model. Do this in three sentences or less. No additional computer output is needed. You are referring to the output in part b.

I would select the six-predictor model containing Adult Mortality, Infant Deaths, Alcohol, Hepatitis B, BMI, and GDP. This model has an R^2 of 0.6646 and a Cp of 8.6454, compared with an R^2 of 0.6651 and a Cp of 8.0326 for the seven-predictor model. Since the six-predictor model explains nearly the same amount of variation with one fewer predictor, it provides a reasonable balance between model fit and simplicity.

Problem 3 (20 points)

Please access the *Ames Housing* Price data. This data set contains 79 explanatory variables describing multiple facets of residential homes in Ames, Iowa, including sale price. We are interested in the following variables. The abbreviated variable names are mostly self-explanatory. They should all be treated as numeric and continuous for our analysis. As always, you will want to make sure JMP brings them in at the appropriate measurement level.

<u>Response</u>

- Sale Price

<u>Regressors</u>

- Total Basmt SF
- 1st Flr SF
- 2nd Flr SF
- Bsmt Full Bath
- Bsmt Half Bath
- Full Bath
- Half Bath
- Bedroom AbvGr
- Kitchen AbvGr
- TotRms AbvGrd
- Wood Deck SF
- Open Porch SF
- Enclosed Porch
- Pool Area

The purpose of this problem is to give you hands-on experience executing a tractable bagging analysis. You will recall from the content video that the concept of bagging is quite broad. So, you are expected to follow the exact simple path we constructed in that video using JMP's Predictive Modeling and Model Screening options.

a. (5 points) We are going to use Analyze -> Predictive Modeling -> Model Screening to do the bagging, but first we all need to construct a validation column, so our answers are uniform.

   To construct the validation column, follow these steps:

   i. Choose Analyze -> Predictive Modeling -> Make Validation Column
   ii. On the overlay that appears, click on the Make Validation Column dropdown on the lower left and choose Make K-Fold Validation Column
   iii. Now put Sale Price in the Y box on the upper right. Do not populate the other boxes.
   iv. Click OK.
   v. On the next overlay change the Number of Folds (K) to 6 and enter 54321 as the Random Seed.
   vi. Click Go.
   vii. This creates a new column on your data table named something like "6 Fold Column."

   To do the bagging with the newly created validation column, follow these steps:

i. Choose Analyze -> Predictive Modeling -> Model Screening back on the data table page.
ii. Put Sale Price in the Y, Response box and the specified regressors in the X, Factor box.
iii. Put the newly created "6 Fold Column" variable (will be character and that is fine) in the Validation box.
iv. Uncheck all boxes under Method except for Fit Least Squares.
v. Check and use 6 for the "K Fold Crossvalidation" entry and for Set Random Seed use 54321.
vi. Run the procedure.

Copy and paste the Training and Validation tables here, with rows ordered by the six Folds.

Training

Method	N	RSquare		RASE	Fold
Fit Least Squares	2197	0.7401	++++++++	40649	1
Fit Least Squares	2197	0.7135	++++++++	42985	2
Fit Least Squares	2197	0.7302	++++++++	40344	3
Fit Least Squares	2197	0.7125	++++++++	42195	4
Fit Least Squares	2196	0.7176	++++++++	42693	5
Fit Least Squares	2196	0.7380	++++++++	40750	6

Validation

Method	N	RSquare		RASE	Fold
Fit Least Squares	243	0.7896	+++++++++	39131	1
Fit Least Squares	243	0.7537	+++++++++	41318	2
Fit Least Squares	243	0.7795	+++++++++	40573	3
Fit Least Squares	243	0.7702	+++++++++	36736	4
Fit Least Squares	244	0.7729	+++++++++	41132	5
Fit Least Squares	244	0.7654	+++++++++	40869	6

b. (5 points) Compute the mean of the RSquare and RASE values across the six folds for the training runs and for the validation runs. Place those values in the table below.

	R^2	**RASE**
	Mean	Mean
Training Set	.7253	41,602.67
Validation Set	.7719	39,959.83

c. (10 points) Refer to the table in part b. In four sentences or less, explain what these values tell you about the possible overfitting of these data.

The results do not suggest substantial overfitting because the validation set has a higher mean R^2 (0.7719) than the training set (0.7253). Additionally, the validation set has a lower mean RASE (39,959.83) compared to the training set (41,602.67), indicating better predictive accuracy on the validation data. If the model were overfitting, we would generally expect a higher R^2 and lower RASE for the training data than for the validation data. Therefore, these results suggest that the model generalizes reasonably well to the validation data.